# A Novel Framework for Mixed Noise Removal From Greenhouse Gases Monitoring Instrument (GMI) Interferogram Images on GF5-02 Satellite

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***Abstract*— To cope with global climate change and monitor the concentration distribution of $CO_2$ and $CH_4$ in the troposphere, the greenhouse gases monitoring instrument (GMI) carried by the GF5-02 satellite was successfully launched on September 7, 2021. GMI utilizes spatial heterodyne spectroscopy techniques to acquire interferogram images. During the acquisition process, interferogram images are inevitably corrupted by various noises, such as defective pixels, baseline drifts and random noise, which will greatly degrade the quality of recovered spectra and further reduce the retrieval accuracy of greenhouse gases. In this article, we model the mixed noise removal of interferogram images as an inverse problem and propose a novel optimization framework by integrating the robust principal component analysis (RPCA), unidirectional total variation (UTV), and penalized least squares (PLS) together. Specifically, the low-rankness of interference fringes and sparsity of defective pixels are both considered in the RPCA model, and the directionality of interference fringes is enhanced by the UTV model, while the PLS model guarantees the fidelity and smoothness of the fitting baselines. An efficient alternating direction method of multipliers procedure is designed to solve the proposed framework. A number of experiments were conducted to illustrate the superior performance of the proposed framework in terms of visual quality and quantitative assessment, as compared with the state-of-the-art algorithms.**

***Index Terms*— Greenhouse gases monitoring instrument (GMI), interferogram, mixed noise, spatial heterodyne spectroscopy (SHS).**

## I. Introduction

A SCIENTIFIC consensus has emerged to mitigate carbon emissions in response to global warming induced by the rise of greenhouse gases (GHGs) such as $CO_2$ and $CH_4$ [1]. Satellite-borne passive remote sensing [2], which measures the absorption spectra of GHGs in the near-infrared and shortwave infrared region, stands out as a key technique for GHG monitoring due to its regular repetitive coverage over large areas. Since the launch of SCIAMACHY in 2002 [3], numerous satellite missions capable of measuring $CO_2$ and/or $CH_4$ concentrations with passive remote sensing detection payloads have been carried out, including OCO-2/3 [4], GOSAT [5], TanSat [6], GF5-01/02 [7], and FY-3D [8].

The greenhouse gas monitoring instrument (GMI) payloads aboard the GF5-01/02 satellites employ an innovative spectroscopic technique known as spatial heterodyne spectroscopy (SHS) to acquire atmospheric absorption spectra [9]. Compared with conventional grating spectroscopy exemplified by OCO-2/3 and Fourier transform spectroscopy (FTS) represented by GOSAT, SHS possesses the advantages of high etendue, ultraspectral resolution, real-time detection, and no moving parts [10], making it especially suitable for quantitative monitoring of atmospheric GHGs.

Essentially, SHS can be regarded as a variant of FTS with a similar process of modulation and demodulation: the radiated light from the target is modulated through the interferometric system to generate an interferogram; the spectrum of the target is subsequently retrieved by performing Fourier transform on the interferogram. One major distinction between SHS and FTS is that an FTS interferogram is sequentially recorded in the time domain by a single detector, whereas SHS utilizes a plane array detector to simultaneously capture the entire 2-D interferogram. In SHS, each pixel behaves as an independent detector, rendering the system highly sensitive to intensity variations across the interferogram [11]. Mixed noise, such as defective pixels (i.e., dead/hot/flicker pixels), pixel-to-pixel nonuniformity, and random noise (e.g. Gaussian/Poisson noise), inevitably results in systematic errors that propagate into spectra. Due to the small spatiotemporal variations in the concentration of GHGs [12], it is crucial for GMI to eliminate the aforementioned nonideal factors in order to achieve high-level detection accuracy.

Before the GMI's launch, a larger number of calibration tests were carried out in the laboratory [13], [14]. Among them, flat-field calibration was performed to remove the nonuniformity effect using the balanced-arm approach [15] and subpixel registration. The bad pixel mask [16] was also measured in the laboratory using integrating sphere to indicate those defective pixels that need to be replaced. However, due to nonideal calibration, variations in space conditions,

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and temporal performance decay, it is imperative to develop and incorporate scenario-based correction algorithms into data processing workflow for on-bit observation data [17]. In [18], an algorithm based on spatial baseline fitting and $3\sigma$-rule was proposed for the detection and correction of defective pixels. On this basis, Han proposed a blind element adaptive correction (BEAC) algorithm [13] to iteratively improve the detection accuracy of defective pixels. To cope with the low-frequency variation of nonuniform intensity distribution, also known as baseline drifts, various detrending algorithms such as polynomial fitting (PF), low-pass filtering [17], wavelet transform (WT) [19], and decomposition techniques [13] (e.g., EMD/VMD) are commonly used. As for suppressing random noise, there is no specific treatment other than row-by-row averaging of recovered spectra to our best knowledge. In practice, the typical data processing procedure for SHS-like systems deals with the mixed noise in a separate manner, without consideration of the mutual coupling and interaction of different types of noise. Furthermore, the significant fluctuations in the central interference fringes will also weaken the capacity of noise correction.

It is noteworthy that the denoising task to tackle mixed noise for remote sensing images has been modeled as an inverse problem and successfully solved by regularization methods [20], [21]. It motivates us to develop a unified variational framework to simultaneously remove mixed noise from the degraded interferogram image. The key idea of this article is to construct regularizations combining robust principal component analysis (RPCA) [22], unidirectional total variation (UTV) [23], and penalized least squares (PLSs) [24] together to constrain the solution. The RPCA model helps to explore the low-rank property of the desired interferogram and suppress defective pixels and random noise as well. The UTV model is designed to retain intrinsic interference fringes and avoid overcorrection. The PLS model is used to maintain the fidelity and smoothness of the fit baseline. The alternating direction method of multipliers (ADMM) is adopted to solve the optimization problem. The proposed framework has been validated by extensive experiments on the four-band GMI interferogram images of the GF5-02 satellite.

The remainder of this article is organized as follows. Section II provides an overview of GMI aboard the GF5-02 satellite, describing its detection principle and level 1 processing procedure. Section III introduces the proposed mixed noise removal framework and the corresponding optimization procedure in detail. In Section IV, comparison experiments are conducted to demonstrate the effectiveness and superiority of the proposed framework. Section V concludes this article.

## II. Overview of the GMI

### A. Detection Principle of GMI

The GMI payload, developed by Anhui Institute of Optics and Fine Mechanics of Chinese Academy of Sciences, was successfully launched into a sun-synchronous orbit at an altitude of 705 km on 7 September 2021. The primary goal of GMI is to quantitatively measure the global concentration distribution of $CO_2$ and $CH_4$ in the troposphere.

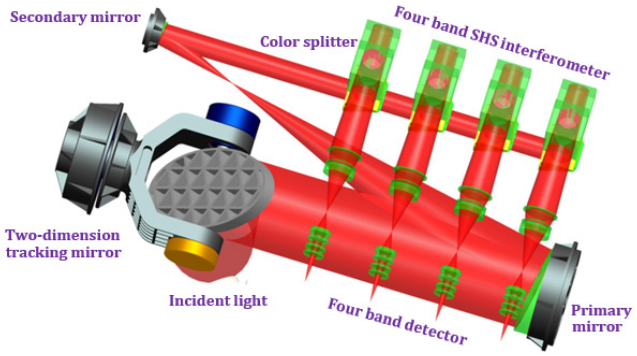



Fig. 1. Optical setup of GMI payload.

The optical setup of GMI is shown in Fig. 1. The incident light is reflected by a 2-D tracking mirror and then enters the Gregory reflecting telescope. Subsequently, the beam diameter is compressed and directed toward the color splitter, where it is divided into four bands: $O_2$ band, $CO_2$-1 band, $CH_4$ band, and $CO_2$-2 band. Each band of the incident light undergoes modulation by an independent SHS interferometer to generate interference fringes. Ultimately, the reimaging lens captures the interference fringes and projects them onto the corresponding detector, resulting in a 2-D interferogram image. The main specifications for each of the four bands are listed in Table I.

The core of GMI is an SHS interferometer comprised of a beam splitter and two diffraction gratings [25], as shown in Fig. 2. Each diffraction grating creates a wavefront at a wavenumber-dependent spatial frequency, separated by a wavenumber-dependent angle for each wavenumber in the incident beam. Specifically, the SHS separates the different wavenumber components of the incident light via the following diffraction grating equation:

$$\sigma(\sin\theta_L + \sin(\theta_L - \gamma)) = m/d \tag{1}$$

where $\sigma$ is the incident wavenumber, $\theta_L$ is the Littrow angle, $\gamma$ is half the angle between the two exiting wavefronts, $m$ is the order of diffraction, and $1/d$ is the density of the grating grooves. When the incident wavenumber happens to be the Littrow wavenumber $\sigma_L$, the diffracted light will return along its original path, that is, $\gamma$ equals zeros. Therefore, the spatial frequency of the interference fringes can be expressed as

$$f(\sigma) = 2\sigma\sin\gamma \approx 4(\sigma - \sigma_L)\tan\theta_L. \tag{2}$$

For the incident spectrum $B(\sigma)$, the interferogram signal recorded by the array detector in the $x$-direction (i.e., grating diffraction direction) can be generally expressed as

$$\begin{aligned} I(x) &= \int_0^\infty B(\sigma)\{1 + \cos 2\pi f(\sigma)x\}d\sigma \\ &= \int_0^\infty B(\sigma)d\sigma + \int_0^\infty B(\sigma)\cos 2\pi f(\sigma)xd\sigma. \end{aligned} \tag{3}$$

According to (3), the measured interferogram consists of nonmodulated term and modulated term (also known as desired interferogram). Hence, the incident spectrum $B(\sigma)$ can be derived through a series of data processing algorithms including Fourier transform.

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TABLE I
MAIN SPECIFICATIONS OF GMI PAYLOAD

| Parameter | Spectral Channel | | | |
|---|---|---|---|---|
| | $O_2$ Band | $CO_2$-1 Band | $CH_4$ Band | $CO_2$-2 Band |
| Spectral range (nm) | 759~769 | 1568~1583 | 1642~1658 | 2043~2058 |
| Spectral resolution ($cm^{-1}$) | 0.6 | 0.27 | 0.27 | 0.27 |
| SNR (albedo=0.3, sun elevation=30°) | 300 | 300 | 250 | 250 |
| Spatial resolution | 10.3×10.3 @ 705km | | | |
| Radiometric calibration | absolution accuracy: 5%; relative accuracy: 2% | | | |
| Observation mode | Nadir: 1,5,7,9 footprints (default: 5 footprints); sun-glint | | | |

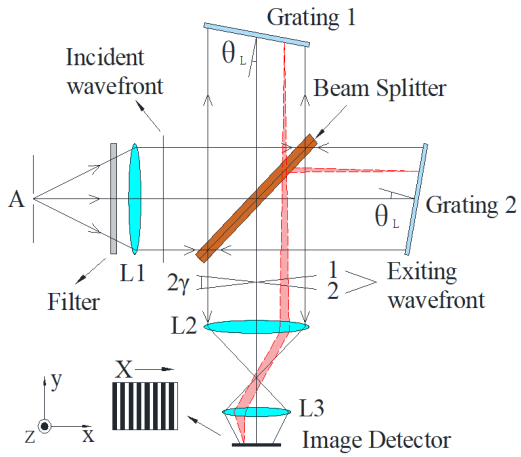



Fig. 2. Structure diagram of SHS interferometer.

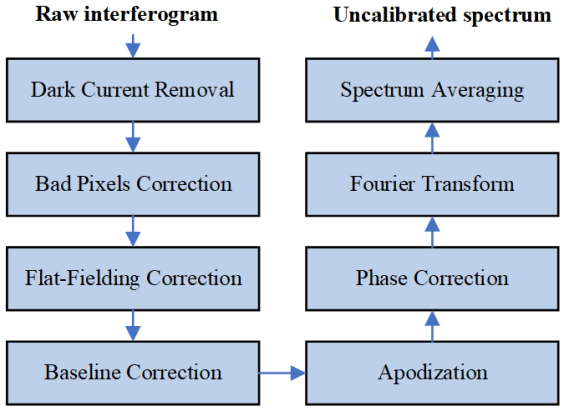



Fig. 3. Flowchart of interferogram processing chain.

### B. Level 1 Processing Procedure

The GMI level 0 data consist of raw SHS interferograms, as well as auxiliary information (e.g., ephemeris data, spacecraft position/attitude data) and laboratory/on-orbit calibration parameters, which are combined to produce geolocated, calibrated radiance spectra by the level 1 processing procedure.

The level 1 processing procedure encompasses geolocation algorithms and data processing algorithms. The geolocation algorithms compute nadir footprint locations (i.e., latitude, longitude, and elevation), as well as associated solar-viewing geometry (i.e., zenith/azimuth angles). The data processing algorithms convert raw interferograms into calibrated spectra through data screening, interferogram processing, and spectral/radiometric calibration. Here, our research is primarily focused on interferogram processing, with additional details on the rest processing available in [17].

As illustrated in Fig. 3, the raw interferograms are corrected for dark current, and bad pixels are real-time identified and replaced through linear interpolation. The flat-field correction and baseline correction are subsequently applied to mitigate nonuniformity and facilitate the extraction of the modulated interferogram. Apodization and phase correction [26] are carried out prior to the Fourier transform to alleviate spectral leakage and asymmetry effect, respectively. Finally, the corrected interferograms with zero-filling are converted to uncalibrated spectra by performing the Fourier transform, which are then averaged row-by-row in order to suppress the random noise.

## III. PROPOSED MIXED NOISE REMOVAL FRAMEWORK

### A. Degradation Model

As noted in the introduction, the measured interferograms are corrupted by various noises, leading to systematic errors in the observed spectra that need to be characterized to perform GHGs retrievals. To achieve a comprehensive view of the various noise, we present the global and local degraded interferogram of GMI $O_2$ band in Fig. 4. The desired interferogram exhibits vertical interference fringes, with the baseline drifts appearing as underlying slow-varying ripples. In addition, the defective pixels are either isolated or aggregated to form lines [see Fig. 4(b)] and speckle patterns [see Fig. 4(c)]. As a result, the interferogram for each individual row exhibits significant variation [see Fig. 4(d) and (e)].

Let $\mathbf{X} \in \mathbb{R}^{M\times N}$ represent the desired interferogram with $M$ and $N$ indicating the number of pixels along the $x$, $y$ directions, respectively, the measured interferogram $\mathbf{Y} \in \mathbb{R}^{M\times N}$ corrupted by baseline drifts $\mathbf{B} \in \mathbb{R}^{M\times N}$, defective pixels $\mathbf{S} \in \mathbb{R}^{M\times N}$, and random noise $\mathbf{G} \in \mathbb{R}^{M\times N}$ can be formulated as

$$\mathbf{Y} = \mathbf{X} + \mathbf{B} + \mathbf{S} + \mathbf{G}. \tag{4}$$

Based on the degradation model (4), the objective of the mixed noise removal is to recover the desired interferogram $\mathbf{X}$ from the degraded interferogram measurements $\mathbf{Y}$. Note that the recovery of $\mathbf{X}$ is a highly ill-posed inverse problem, it is essential to incorporate the prior information of the desired interferogram and mixed noise to constrain the solution.

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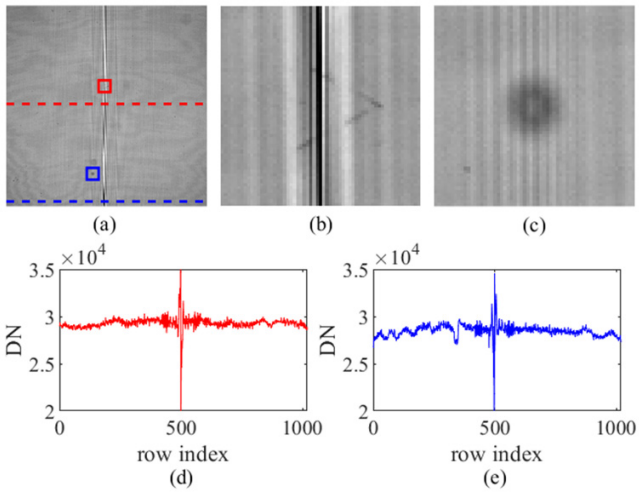



Fig. 4. Illustration of the forms of various noise in the GMI $O_2$ band interferogram. (a) Measured interferogram. (b) and (c) Zoomed-in areas corresponding to the red and blue boxes in (a), respectively. (d) and (e) Interferogram curves corresponding to the red and blue dotted lines in (a), respectively.

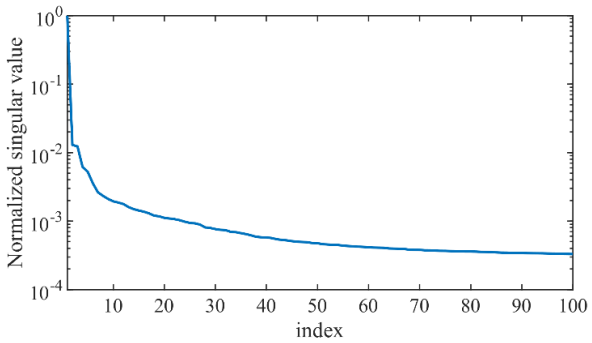



Fig. 5. Singular values corresponding to the observed interferogram image of GMI $O_2$ band.

### B. Prior Characteristics Analysis

*1) Low-Rankness Property of Desired Interferogram:* According to (3), it can be seen that each row of an ideal desired interferogram would be identical, indicating the rank-1 property of $\mathbf{X}$. Despite the presence of nonideal factors such as phase distortion and nonuniformity, which may potentially increase the rank, the desired interferogram $\mathbf{X}$ is still expected to be of low rank, that is, $\text{rank}(\mathbf{X}) = r \ll \min(M, N)$.

The property of low-rankness also exists in the measured interferogram, which typically contains full-rank noises. As depicted in Fig. 5, the normalized singular values of the measured interferogram exhibit a rapid decay with most values falling below $10^{-2}$, indicating that the desired interferogram $\mathbf{X}$ has a low-rank structure.

Furthermore, the matrix $\mathbf{S}$ can be considered as a bad pixel mask with only a minority of pixels having nonzero values. In other words, the matrix $\mathbf{S}$ exhibits a sparse property. By combining the low-rank property of $\mathbf{X}$ and sparse priori of $\mathbf{S}$, Wright and Candès proposed an RPCA model [22] with the following formulation:

$$\min_{\mathbf{X},\mathbf{S}} \text{rank}(\mathbf{X}) + \lambda \|\mathbf{S}\|_0, \quad \text{s.t. } \mathbf{Y} = \mathbf{X} + \mathbf{S} \tag{5}$$

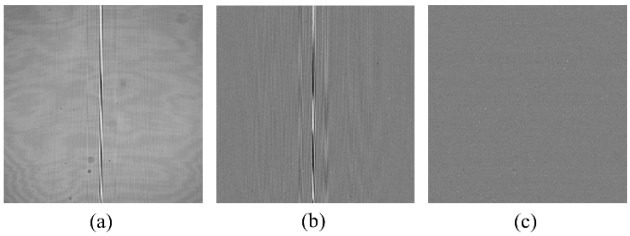



Fig. 6. Directional characteristic of the interferogram image. (a) Observed interferogram image of GMI $O_2$ band. (b) Horizontal gradient. (c) Vertical gradient.

where $\lambda$ is a regularization parameter used to balance the contribution of low-rank norm and $l_0$-norm.

Unfortunately, the model (5) is a highly nonconvex optimization problem with no efficient solution. A tractable approach is to relax the rank function with nuclear norm and substitute the $l_0$-norm with $l_1$-norm, resulting in the following convex optimization problem:

$$\min_{\mathbf{X},\mathbf{S}} \|\mathbf{X}\|_* + \lambda \|\mathbf{S}\|_1, \quad \text{s.t. } \mathbf{Y} = \mathbf{X} + \mathbf{S}, \quad \text{rank}(\mathbf{X}) \le r \tag{6}$$

where $\|\cdot\|_*$ stands for the nuclear norm of a matrix that is the sum of its singular values and the $l_1$-norm $\|\cdot\|_1$ represents the sum of the absolute value of all entries.

*2) Unidirectional Property of Desired Interferogram:* The interferogram image exhibits a prominent directional signature, manifesting noticeable variations primarily concentrated across the interference fringes. As shown in Fig. 6, the horizontal gradient across the fringes is severely affected by the fringes, whereas the gradient along the fringes is barely influenced. This motivates us to suppress the gradient along the fringes while preserving the gradient across the fringes. To achieve this, one can utilize UTV model, which was initially designed for remote sensing image destriping [23].

However, in contrast to the UTV model used in traditional destring tasks, the fringes of interferogram are considered as desired signals containing valuable modulation information, rather than stripe noise that need to be removed. Therefore, the UTV model is modified as

$$\mathcal{J}_{\text{UTV}}(\mathbf{X}) = \gamma \|\nabla_v \mathbf{X}\|_1 + \eta \|\nabla_h (\mathbf{Y} - \mathbf{X})\|_1 \tag{7}$$

where $\nabla_v$ and $\nabla_h$ denote the linear differential operator in the vertical (along-fringe) and horizontal (across-fringe) direction, respectively, and $\gamma$, $\eta$ are regularization parameters to tradeoff the two terms.

*3) PLS Model for Baseline Estimation:* The concept of PLS model was initially developed to effectively smooth 1-D equally spaced data [24]. Consider an interferogram sequence $\boldsymbol{y}$ of length $N$, extracted from the observed interferogram image. The principle underlying the PLS model is that the baseline signal $\boldsymbol{z}$ should follow the trend of $\boldsymbol{y}$ while maintaining its smoothness by minimizing the following objective function:

$$Q = (\boldsymbol{y} - \boldsymbol{z})^T (\boldsymbol{y} - \boldsymbol{z}) + \lambda (\mathbf{D}\boldsymbol{z})^T \mathbf{D}\boldsymbol{z} \tag{8}$$

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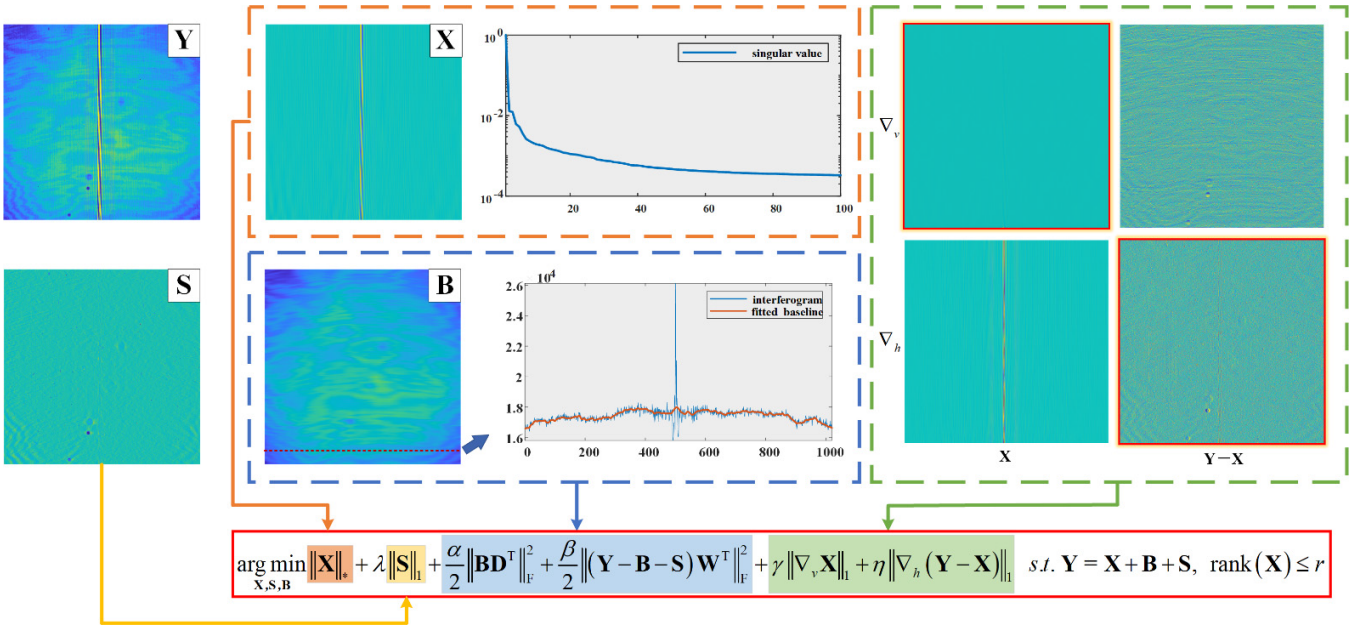



Fig. 7. Flowchart describing the proposed mixed noise removal framework.

where $\mathbf{D}$ is the 2nd-order difference matrix given by

$$\mathbf{D} = \begin{bmatrix} 1 & -2 & 1 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 1 & -2 & 1 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 1 & -2 & 1 \end{bmatrix} \in \mathbb{R}^{(N-2)\times N}. \tag{9}$$

The first term in (8) represents the fidelity of $z$ to data $y$, whereas the second term stands for the smoothness of $z$. The balance between two terms is controlled by the parameter $\lambda$.

To accommodate the presence of defective pixels, a weight vector $\boldsymbol{w}$ is introduced (with $\boldsymbol{w}_i = 0$ for defective pixels and $\boldsymbol{w}_i = 1$ otherwise) into an asymmetric least squares model

$$Q = (\boldsymbol{y} - \boldsymbol{z})^T\mathbf{W}(\boldsymbol{y} - \boldsymbol{z}) + \lambda(\mathbf{D}\boldsymbol{z})^T\mathbf{D}\boldsymbol{z} \tag{10}$$

where $\mathbf{W}$ is a diagonal matrix with weight vector $\boldsymbol{w}$ on its diagonal.

For the baseline estimation of interferogram, the defective pixels are characterized as sparse noise, which is treated separately. Hence, we modify the PLS model and rewrite it in a matrix form

$$\mathcal{J}_{\mathrm{PLS}}(\mathbf{X}) = \frac{\alpha}{2}\left\|\mathbf{B}\mathbf{D}^T\right\|_F^2 + \frac{\beta}{2}\left\|(\mathbf{Y}-\mathbf{B}-\mathbf{S})\mathbf{W}^T\right\|_F^2 \tag{11}$$

where $\alpha$ and $\beta$ are regularization parameters to balance the two terms. The first term penalizes the overall roughness of $\mathbf{B}$ in the $x$-direction, whereas the second term ensures the fidelity of $\mathbf{B}$ to the measured interferogram with $\mathbf{S}$ excluded. It is worth noting that the weight matrix $\mathbf{W}$ is still retained in (11) to prevent overfitting of the baselines in the center-burst region. For this reason, the diagonal of $\mathbf{W}$ shall be set to zero in the vicinity of the center-burst position.

### C. Proposed Framework

Based on the forementioned analysis, the optimization model proposed in this article for mixed noise removal can be summarized as follows:

$$\begin{aligned} \arg\min_{\mathbf{X},\mathbf{S},\mathbf{B}} \ & \|\mathbf{X}\|_* + \gamma\|\nabla_v\mathbf{X}\|_1 + \eta\|\nabla_h(\mathbf{Y}-\mathbf{X})\|_1 + \lambda\|\mathbf{S}\|_1 \\ & + \frac{\alpha}{2}\left\|\mathbf{B}\mathbf{D}^T\right\|_{\mathbf{F}}^2 + \frac{\beta}{2}\left\|(\mathbf{Y}-\mathbf{B}-\mathbf{S})\mathbf{W}^T\right\|_{\mathbf{F}}^2 \\ \text{s.t. } & \mathbf{Y} = \mathbf{X}+\mathbf{B}+\mathbf{S}, \quad \operatorname{rank}(\mathbf{X}) \le r \end{aligned} \tag{12}$$

where $\gamma$, $\eta$, $\lambda$, $\alpha$, and $\beta$ are the regularization parameters, and $r$ denotes the upper bound of the rank of $\mathbf{X}$. In (12), the RPCA, UTV, and PLS models are integrated into a unified framework for mixed noise removal, as illustrated in Fig. 7.

### D. Optimization Procedure

By introducing four auxiliary variables $\mathbf{Z}$, $\mathbf{T}$, $\mathbf{U}$, and $\mathbf{V}$, we rewrite (12) into the following equivalent minimization problem:

$$\begin{aligned} \arg\min_{\mathbf{X},\mathbf{S},\mathbf{B},\mathbf{Z},\mathbf{T},\mathbf{U},\mathbf{V}} \ & \|\mathbf{Z}\|_* + \gamma\|\mathbf{U}\|_1 + \eta\|\mathbf{V}\|_1 + \lambda\|\mathbf{T}\|_1 \\ & + \frac{\alpha}{2}\left\|\mathbf{B}\mathbf{D}^T\right\|_{\mathbf{F}}^2 + \frac{\beta}{2}\left\|(\mathbf{Y}-\mathbf{B}-\mathbf{S})\mathbf{W}^T\right\|_{\mathbf{F}}^2 \\ \text{s.t. } & \mathbf{Y} = \mathbf{X}+\mathbf{S}+\mathbf{B}, \quad \mathbf{Z} = \mathbf{X}, \quad \mathbf{T} = \mathbf{S} \\ & \mathbf{U} = \nabla_v\mathbf{X}, \quad \mathbf{V} = \nabla_h(\mathbf{Y}-\mathbf{X}), \quad \operatorname{rank}(\mathbf{Z}) \le r \end{aligned} \tag{13}$$

with an associated augmented Lagrangian function

$$\begin{aligned} & \mathcal{L}(\mathbf{X},\mathbf{S},\mathbf{B},\mathbf{Z},\mathbf{T},\mathbf{U},\mathbf{V}) \\ & = \|\mathbf{Z}\|_* + \gamma\|\mathbf{U}\|_1 + \eta\|\mathbf{V}\|_1 + \lambda\|\mathbf{T}\|_1 + \frac{\alpha}{2}\left\|\mathbf{B}\mathbf{D}^T\right\|_F^2 \\ & \quad + \frac{\beta}{2}\left\|(\mathbf{Y}-\mathbf{B}-\mathbf{S})\mathbf{W}^T\right\|_F^2 + \langle\boldsymbol{\Lambda}_1, \mathbf{Y}-\mathbf{X}-\mathbf{B}-\mathbf{S}\rangle \\ & \quad + \langle\boldsymbol{\Lambda}_2, \mathbf{Z}-\mathbf{X}\rangle + \langle\boldsymbol{\Lambda}_3, \mathbf{T}-\mathbf{S}\rangle + \langle\boldsymbol{\Lambda}_4, \mathbf{U}-\nabla_v\mathbf{X}\rangle \\ & \quad + \langle\boldsymbol{\Lambda}_5, \mathbf{V}-\nabla_h(\mathbf{Y}-\mathbf{X})\rangle \\ & \quad + \frac{\mu}{2}\left(\|\mathbf{Y}-\mathbf{X}-\mathbf{B}-\mathbf{S}\|_F^2 + \|\mathbf{Z}-\mathbf{X}\|_F^2 + \|\mathbf{T}-\mathbf{S}\|_F^2\right. \\ & \qquad \left. + \|\mathbf{U}-\nabla_v\mathbf{X}\|_F^2 + \|\mathbf{V}-\nabla_h(\mathbf{Y}-\mathbf{X})\|_F^2\right) \\ & \text{s.t. } \operatorname{rank}(\mathbf{Z}) \le r \end{aligned} \tag{14}$$

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where the penalty parameter $\mu$ can be adjusted regularly to accelerate convergence and $\mathbf{\Lambda}_i$ $(i = 1, 2, 3, 4, 5)$ denote the Lagrangian multipliers.

For solving the multivariable optimization problem, a natural way is to alternately and iteratively optimize the augmented Lagrangian function (14) over one variable while fixing the others. Specifically, in the $(k+1)$th iteration, the variables involved in the model (13) can be updated by solving the following subproblems within the ADMM framework.

*1) $\mathbf{X}$-Related Subproblem:*

$$\begin{aligned}\mathbf{X}^{(k+1)} = \arg\min_{\mathbf{X}} &\left\|\mathbf{Y}-\mathbf{X}-\mathbf{B}^{(k)}-\mathbf{S}^{(k)}+\mathbf{\Lambda}_1^{(k)}\Big/\mu\right\|_F^2 \\ &+\left\|\mathbf{Z}^{(k)}-\mathbf{X}+\mathbf{\Lambda}_2^{(k)}\Big/\mu\right\|_F^2 \\ &+\left\|\mathbf{U}^{(k)}-\nabla_v\mathbf{X}+\mathbf{\Lambda}_4^{(k)}\Big/\mu\right\|_F^2 \\ &+\left\|\mathbf{V}^{(k)}-\nabla_h(\mathbf{Y}-\mathbf{X})+\mathbf{\Lambda}_5^{(k)}\Big/\mu\right\|_F^2. \end{aligned} \tag{15}$$

As a convex optimization model, this problem can be solved using the following linear system:

$$\begin{aligned}&\left(2\mathbf{I}+\nabla_h^T\nabla_h+\nabla_v^T\nabla_v\right)\mathbf{X}^{(k+1)} \\ &\quad=\nabla_h^T\left(\nabla_h\mathbf{Y}-\mathbf{V}^{(k)}-\mathbf{\Lambda}_5^{(k)}\Big/\mu\right)+\nabla_v^T\left(\mathbf{U}^{(k)}+\mathbf{\Lambda}_4^{(k)}\Big/\mu\right) \\ &\qquad+\mathbf{Y}+\mathbf{Z}^{(k)}-\mathbf{B}^{(k)}-\mathbf{S}^{(k)}+\left(\mathbf{\Lambda}_1^{(k)}+\mathbf{\Lambda}_2^{(k)}\right)\Big/\mu. \end{aligned} \tag{16}$$

Thanks to the block-circulant structure of the matrix corresponding to the operators $\nabla_h^T\nabla_h$ and $\nabla_v^T\nabla_v$, it can be diagonalized by the 2-D FFT matrix. Hence, there is a closed-form solution

$$\mathbf{X}^{(k+1)} = \mathcal{F}^{-1}\left(\frac{\mathcal{F}(\mathbf{H})}{\mathbf{2}+|\mathcal{F}(\nabla_h)|^2+|\mathcal{F}(\nabla_v)|^2}\right) \tag{17}$$

where $\mathbf{H} = \nabla_h^T(\nabla_h\mathbf{Y}-\mathbf{V}^{(k)}-\mathbf{\Lambda}_5^{(k)}/\mu)+\nabla_v^T(\mathbf{U}^{(k)}+\mathbf{\Lambda}_4^{(k)}/\mu)+\mathbf{Y}+\mathbf{Z}^{(k)}-\mathbf{B}^{(k)}-\mathbf{S}^{(k)}+(\mathbf{\Lambda}_1^{(k)}+\mathbf{\Lambda}_2^{(k)})/\mu$, $\mathcal{F}(\cdot)$ and $\mathcal{F}^{-1}(\cdot)$ are the fast 2-D Fourier transform operator and its inverse operator, respectively, $|\cdot|^2$ is the element-wise square, and the division is also performed element-wisely.

*2) $\mathbf{S}$-Related Subproblem:*

$$\begin{aligned}\mathbf{S}^{(k+1)} = \arg\min_{\mathbf{S}} &\frac{\beta}{2}\left\|\left(\mathbf{Y}-\mathbf{B}^{(k)}-\mathbf{S}\right)\mathbf{W}^T\right\|_F^2 \\ &+\frac{\mu}{2}\left(\left\|\mathbf{T}^{(k)}-\mathbf{S}+\mathbf{\Lambda}_3^{(k)}\Big/\mu\right\|_F^2\right. \\ &\left.+\left\|\mathbf{Y}-\mathbf{X}^{(k+1)}-\mathbf{B}^{(k)}-\mathbf{S}+\mathbf{\Lambda}_1^{(k)}\Big/\mu\right\|_F^2\right). \end{aligned} \tag{18}$$

This is a quadratic minimization problem, which has the following closed-form solution:

$$\begin{aligned}\mathbf{S}^{(k+1)} = &\left[\mu\left(\mathbf{Y}-\mathbf{X}^{(k+1)}-\mathbf{B}^{(k)}+\mathbf{T}^{(k)}\right)+\mathbf{\Lambda}_1^{(k)}+\mathbf{\Lambda}_3^{(k)}\right. \\ &\left.+\beta\left(\mathbf{Y}-\mathbf{B}^{(k)}\right)\mathbf{W}^T\mathbf{W}\right]\left(\beta\mathbf{W}^T\mathbf{W}+2\mu\mathbf{I}\right)^{-1}. \end{aligned} \tag{19}$$

*3) $\mathbf{B}$-Related Subproblem:*

$$\begin{aligned}\mathbf{B}^{(k+1)} = \arg\min_{\mathbf{B}} &\frac{\alpha}{2}\left\|\mathbf{B}\mathbf{D}^T\right\|_F^2+\frac{\beta}{2}\left\|\left(\mathbf{Y}-\mathbf{B}-\mathbf{S}^{(k+1)}\right)\mathbf{W}^T\right\|_F^2 \\ &+\frac{\mu}{2}\left\|\mathbf{Y}-\mathbf{X}^{(k+1)}-\mathbf{B}-\mathbf{S}^{(k+1)}+\mathbf{\Lambda}_1^{(k)}\Big/\mu\right\|_F^2. \end{aligned} \tag{20}$$

Similarly, we can obtain its closed-form solution

$$\begin{aligned}\mathbf{B}^{(k+1)} = &\left[\beta\left(\mathbf{Y}-\mathbf{S}^{(k+1)}\right)\mathbf{W}^T\mathbf{W}+\mu\left(\mathbf{Y}-\mathbf{X}^{(k+1)}-\mathbf{S}^{(k+1)}\right)\right. \\ &\left.+\mathbf{\Lambda}_1^{(k)}\right]\left(\alpha\mathbf{D}^T\mathbf{D}+\beta\mathbf{W}^T\mathbf{W}+\mu\mathbf{I}\right)^{-1}. \end{aligned} \tag{21}$$

*4) $\mathbf{Z}$-Related Subproblem:*

$$\mathbf{Z}^{(k+1)} = \arg\min_{\operatorname{rank}(\mathbf{Z})\le r}\|\mathbf{Z}\|_* + \frac{\mu}{2}\left\|\mathbf{Z}-\mathbf{X}^{(k+1)}+\mathbf{\Lambda}_2^{(k)}\Big/\mu\right\|_{\mathbf{F}}^2. \tag{22}$$

By utilizing the singular value decomposition (SVD) of the matrix and implementing a soft-thresholding operation element-wisely, it can be easily deduced that

$$\mathbf{Z}^{(k+1)} = \mathcal{D}_{1/\mu}\left(\mathbf{X}^{(k+1)}-\mathbf{\Lambda}_2^{(k)}\Big/\mu\right) \tag{23}$$

where

$$\begin{aligned}\mathcal{D}_\delta\left(\tilde{\mathbf{W}}\right) &= \tilde{\mathbf{U}}\mathcal{D}_\delta(\mathbf{\Sigma}_r)\tilde{\mathbf{V}}^T \\ &= \tilde{\mathbf{U}}\operatorname{diag}\{\max(\sigma_i-\delta,0)\}_{1\le i\le r}\tilde{\mathbf{V}}^T. \end{aligned} \tag{24}$$

$\tilde{\mathbf{U}}\tilde{\mathbf{\Sigma}}\tilde{\mathbf{V}}^T$ is the SVD of the given matrix $\tilde{\mathbf{W}}$ and $\{\sigma_i\}_{1\le i\le r}$ are the first $r$ singular values of $\tilde{\mathbf{W}}$ in descending order.

*5) $\mathbf{T}$-Related Subproblem:*

$$\mathbf{T}^{(k+1)} = \arg\min_{\mathbf{T}}\lambda\|\mathbf{T}\|_1+\frac{\mu}{2}\left\|\mathbf{T}-\mathbf{S}^{(k+1)}+\mathbf{\Lambda}_3^{(k)}\Big/\mu\right\|_F^2. \tag{25}$$

By introducing the following soft-thresholding operator:

$$\mathcal{R}_\Delta(x) = \operatorname{sign}(x)\cdot\max(|x|-\Delta,0) \tag{26}$$

where $x\in\mathbb{R}$ and $\Delta>0$, the optimization of (25) can be expressed as follows:

$$\mathbf{T}^{(k+1)} = \mathcal{R}_{\lambda/\mu}\left(\mathbf{S}^{(k+1)}-\mathbf{\Lambda}_3^{(k)}\Big/\mu\right). \tag{27}$$

*6) $\mathbf{U}$-Related Subproblem:*

$$\mathbf{U}^{(k+1)} = \arg\min_{\mathbf{U}}\gamma\|\mathbf{U}\|_1+\frac{\mu}{2}\left\|\mathbf{U}-\nabla_v\mathbf{X}^{(k+1)}+\mathbf{\Lambda}_4^{(k)}\Big/\mu\right\|_F^2. \tag{28}$$

This problem is similar to the optimization related to $\mathbf{T}$, with the solution expressed as

$$\mathbf{U}^{(k+1)} = \mathcal{R}_{\gamma/\mu}\left(\nabla_v\mathbf{X}^{(k+1)}-\mathbf{\Lambda}_4^{(k)}\Big/\mu\right). \tag{29}$$

*7) $\mathbf{V}$-Related Subproblem:*

$$\begin{aligned}\mathbf{V}^{(k+1)} = \arg\min_{\mathbf{V}}&\eta\|\mathbf{V}\|_1 \\ &+\frac{\mu}{2}\left\|\mathbf{V}-\nabla_h\left(\mathbf{Y}-\mathbf{X}^{(k+1)}\right)+\mathbf{\Lambda}_5^{(k)}\Big/\mu\right\|_F^2. \end{aligned} \tag{30}$$

Similarly, the optimization result is given by

$$\mathbf{V}^{(k+1)} = \mathcal{R}_{\eta/\mu}\left(\nabla_h\left(\mathbf{Y}-\mathbf{X}^{(k+1)}\right)-\mathbf{\Lambda}_5^{(k)}\Big/\mu\right). \tag{31}$$

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*8) Update the Lagrangian Multipliers:*

$$\begin{cases} \mathbf{\Lambda}_1^{(k+1)} = \mathbf{\Lambda}_1^{(k)} + \mu\left(\mathbf{Y} - \mathbf{X}^{(k+1)} - \mathbf{B}^{(k+1)} - \mathbf{S}^{(k+1)}\right) \\ \mathbf{\Lambda}_2^{(k+1)} = \mathbf{\Lambda}_2^{(k)} + \mu\left(\mathbf{Z}^{(k+1)} - \mathbf{X}^{(k+1)}\right) \\ \mathbf{\Lambda}_3^{(k+1)} = \mathbf{\Lambda}_3^{(k)} + \mu\left(\mathbf{T}^{(k+1)} - \mathbf{S}^{(k+1)}\right) \\ \mathbf{\Lambda}_4^{(k+1)} = \mathbf{\Lambda}_4^{(k)} + \mu\left(\mathbf{U}^{(k+1)} - \nabla_v \mathbf{X}^{(k+1)}\right) \\ \mathbf{\Lambda}_5^{(k+1)} = \mathbf{\Lambda}_5^{(k)} + \mu\left(\mathbf{V}^{(k+1)} - \nabla_h\left(\mathbf{Y} - \mathbf{X}^{(k+1)}\right)\right). \end{cases} \tag{32}$$

In summary, based on the aforementioned description, an extended ADMM method for the proposed framework can be designed to solve the optimization problem, as described in Algorithm 1.

**Algorithm 1** The Proposed Mixed Noise Removal Framework

Input: Original observed interferogram image $\mathbf{Y} \in \mathbb{R}_{M\times N}$, desired rank $r$, regularized parameters $\lambda$, $\alpha$, $\beta$, $\gamma$, $\eta$, weight matrix $\mathbf{W}$, stopping criteria $\varepsilon_i$ $(i = 1, 2, 3, 4, 5)$.

Output: Recovered interferogram image $\mathbf{X}$

Initialize: $\mathbf{B}(:, j) = \text{mean}(\mathbf{Y}(:, j))$, $1 \le j \le N$; $\mathbf{X} = \mathbf{Z} = \mathbf{Y} - \mathbf{B}$, $\mathbf{S} = \mathbf{T} = \mathbf{U} = \mathbf{V} = \mathbf{0}$, $\mathbf{\Lambda}_i = \mathbf{0}(i = 1, 2, 3, 4, 5)$, $\mu = 10^{-2}$, $\mu_{\max} = 10^6$, $\rho = 1.5$ and $k = 0$

Repeat until convergence

1) Update $\mathbf{X}^{(k+1)}, \mathbf{S}^{(k+1)}, \mathbf{B}^{(k+1)}, \mathbf{Z}^{(k+1)}, \mathbf{T}^{(k+1)}, \mathbf{U}^{(k+1)}, \mathbf{V}^{(k+1)}$ via (17), (19), (21), (23), (27), (29) and (30), respectively;
2) Update Lagrangian multipliers $\mathbf{\Lambda}_1^{(k+1)}$, $\mathbf{\Lambda}_2^{(k+1)}$, $\mathbf{\Lambda}_3^{(k+1)}$, $\mathbf{\Lambda}_4^{(k+1)}$ and $\mathbf{\Lambda}_5^{(k+1)}$ via (32);
3) Update the parameter $\mu := \min(\rho\mu, \mu_{\max})$;
4) Check the convergence conditions:
$\|\mathbf{Y} - \mathbf{X}^{(k+1)} - \mathbf{B}^{(k+1)} - \mathbf{S}^{(k+1)}\|_F / \|\mathbf{Y}\|_F \le \varepsilon_1$,
$\|\mathbf{Z}^{(k+1)} - \mathbf{X}^{(k+1)}\|_\infty \le \varepsilon_2$, $\|\mathbf{T}^{(k+1)} - \mathbf{S}^{(k+1)}\|_\infty \le \varepsilon_3$,
$\|\mathbf{U}^{(k+1)} - \nabla_v \mathbf{X}^{(k+1)}\|_\infty \le \varepsilon_4$, $\|\mathbf{V}^{(k+1)} - \nabla_h(\mathbf{Y} - \mathbf{X}^{(k+1)})\|_\infty \le \varepsilon_5$

### *E. Computing Complexity Analysis*

According to the previous section, the computing complexity of Algorithm 1 can be evaluated from different subproblems: 1) the $\mathbf{X}$-related subproblem could be accelerated by using the FFT technique with a computing complexity of $\mathcal{O}(MN\log(MN))$; 2) for the $\mathbf{S}$-, $\mathbf{B}$-related subproblems, the corresponding closed-form solutions have the computing complexity of $\mathcal{O}(MN)$; 3) for the $\mathbf{Z}$-related subproblem, the SVD operation is employed with a computing complexity of $\mathcal{O}(MNr)$; and 4) for the $\mathbf{T}$-, $\mathbf{U}$-, $\mathbf{V}$-related subproblems, the computing complexity that depends on a soft shrinkage operator is $\mathcal{O}(MN)$. Taking all parts into consideration, the computing complexity of the entire process is $\mathcal{O}(MN\log(MN) + MNr)$.

## IV. Experimental Results and Discussions

To validate the efficacy of the proposed framework, a series of experiments were carried out from three perspectives: defective pixel removal performance, baseline drift correction performance, and comprehensive effect on the restored spectra. In Section IV-A, the results of defective pixel removal are presented and compared with the threshold-based BEAC algorithm, commonly utilized in the level-1 processing of GMI. In Section IV-B, the baseline correction performance is compared with several state-of-the-art methods including PF, Butterworth low-pass filtering (BLPF), empirical mode decomposition (EMD), and WT. In Section IV-C, the quantitative evaluation of the radiance spectra is conducted. For fair comparison, all these algorithms are tested under identical conditions. In addition, other steps in the L1 processing procedure (see Fig. 3) remain consistent, including apodization, phase correction, and spectral calibration.

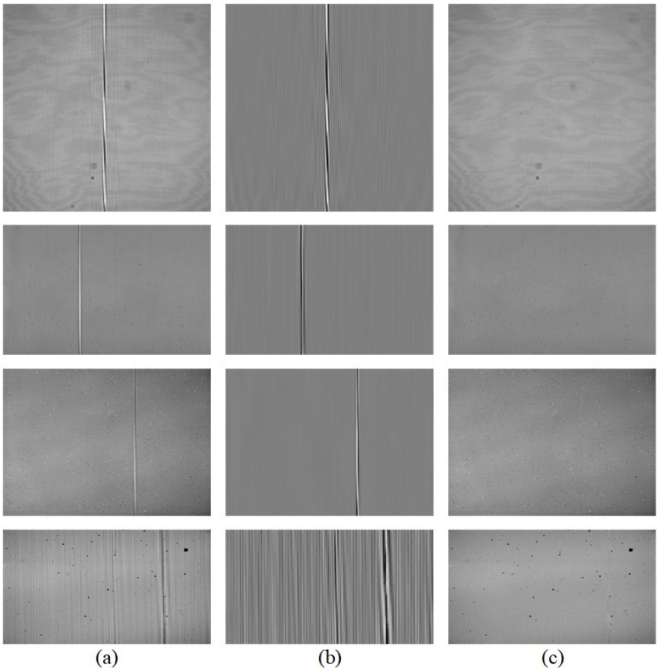



Fig. 8. (Top to Bottom) The visual results of corrected interferogram using the proposed method on $O_2$, $CO_2$-1, $CH_4$, and $CO_2$-2 bands, respectively. (a) Degraded interferogram image. (b) Desired interferogram image. (c) Residual noise.

The experiments were carried out utilizing the four-band interferograms derived from GMI's atmospheric observation data at 09:43:39 on 11 November 2021, with associated nadir footprint located at a latitude of 10.979° and a longitude of 21.743°. In the implementation, the stopping criteria are set as $10^{-4}$, the weight matrix $\mathbf{W}$ is adjusted according to the center-burst position of each band. The determination of desired rank and regularization parameters, along with convergence analysis, will be discussed in Section IV-D. In addition, the parameter settings of those competing algorithms are manually fine-tuned to the optimal.

Fig. 8 depicts the correction results of the proposed method on four-band interferograms, which intuitively indicates its good performance in effectively removing mixed noise. A comparative analysis will be presented in the subsequent subsections.

### *A. Performance of Defective Pixel Removal*

After performing Fourier transformation, a defective pixel will introduce a cosine oscillation into the spectral domain with the frequency proportional to the corresponding optical path difference (OPD). As demonstrated in previous

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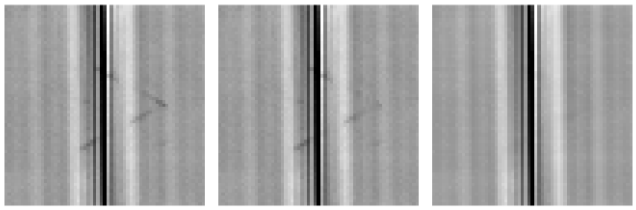



Fig. 9. Comparison results of defective pixel removal on $O_2$ band interferogram image. (a) Subimage of observed interferogram image corresponding to Fig. 4(b). (b) BEAC. (c) Proposed method.

research [27], a defective pixel located near the zero OPD position will bring in greater spectral distortion. Therefore, we should pay more attention to the removal performance of defective pixels located near the zero OPD position.

Fig. 9 illustrates the comparison results for defective pixel removal using BEAC and the proposed method. It is evident from Fig. 9(a) that two intersecting line segments are present in the center-burst region. The BEAC method fails to remove them thoroughly, primarily attributed to the slight inclination of the interferogram caused by relative rotation of gratings. The standard deviation along the spatial dimension in the center-burst region could be so large that the defective pixels are almost indistinguishable. The proposed method uses RPCA model to simultaneously exploit the low-rankness of the desired interferogram and sparsity of the defective pixels, so that the defective pixels especially in the center-burst region can be separated well.

Without loss of generality, we present the results of defective pixel removal on four-band interferograms in Fig. 10. Obviously, the defective pixels of four bands are reduced to almost none using the proposed method, while the results of the BEAC method still contain some residual defective pixels. This phenomenon further demonstrates the superiority of the proposed method in removing defective pixels.

### *B. Performance of Baseline Correction*

A baseline drift acts as a slow-varying trend, leading to spectral distortion in the low-frequency region. Generally, a good baseline correction method should eliminate the baseline drifts while maintaining the interference fringes. In other words, the separated baseline should not contain interference fringes and vice versa.

Fig. 11 visually compares the baseline correction performance of the proposed method with that of other competing methods. It can be clearly seen that the proposed method performs the best, effectively eliminating the baseline drifts while preserving the interference fringes well. PF preserves the fringes well, but also introduces artifacts (e.g., the ripples in the bottom left). Reversely, WT removes baseline drifts more completely, however, some interference fringes in the center-burst region are lost. The performance of BLPF and EMD lie somewhere between PF and WT. More specifically, PF uses a polynomial curve as the fit baseline, so the transition between the center-burst region is much smoother. Due to the filtering behavior of BLPF, EMD, and WT methods, it is difficult to completely decompose the interference fringes in the center-burst region into just a few frequency components. For this phenomenon, we introduce a weight matrix $\mathbf{W}$ to prevent overfitting in the center-burst region while keeping the smoothness of baseline.

Furthermore, the synthetic results of defective pixel removal and baseline correction for the speckle with certain size [see Fig. 4(c)] are shown in the enlarged subregion in Fig. 11 (delineated in red boxes). The speckle can be viewed as continuously distributed defective pixels or a baseline with a narrow waveform, neither of which can be easily handled by existing defective pixel removal or baseline correction methods. Notably, the proposed method can separate the speckle with vivid edges from the degraded interferogram without corrupting the fringes. The main reason is that the proposed method benefits from the unified optimization framework for both defective pixels and baseline drifts.

Another significant difference between the proposed method and other competing baseline correction methods is that the competing methods are handled in a row-by-row manner, so the same parameter settings may not work well for every row. The proposed method introduces a PLS model to minimize the overall correction error contributed by each row, thus achieving global optimal results. Figs. 12 and 13 present the baseline correction results of two interferogram curves (corresponding to Fig. 4(d) and (e), respectively) using different methods. It is clear that the baseline in Fig. 13 is more complex and contaminated by speckle noise. As a result, the performance of PF degrades, and so do BLPF and EMD. Among the competing methods, WT performs comparatively better, although over-fitting occurs in the center-burst region. In contrast, the proposed method can achieve almost identical satisfactory results in both cases, indicating the effectiveness and robustness of our method for simultaneously removing the defective pixels and baseline drifts.

### *C. Comprehensive Effect on the Recovered Spectra*

Since the inversion of atmospheric GHGs requires high-precision radiance spectra, it is necessary to evaluate the performance of the proposed method in terms of recovered spectra. The recovered spectra can be obtained from the corrected interferogram following two steps. The corrected interferogram is first converted into spectrogram as shown in Fig. 14 after row-by-row phase correction and Fourier transform. Then, the digital value of the averaged spectrum is converted into the radiance value with physical significance through the application of spectral and radiometric calibration. The final four bands of calibrated radiance spectrum are illustrated in Fig. 15.

In Fig. 14, we also compare the spectrogram from corrected interferogram of BEAC + PF method and the proposed method. It is clear that the noise in the spectrogram of the proposed method has been evidently reduced, mainly due to the suppression of random noise in the interferogram image. Meanwhile, the spectral distortion between adjacent rows, which is the interaction result of residual mixed noise and phase correction processing, is effectively alleviated as well.

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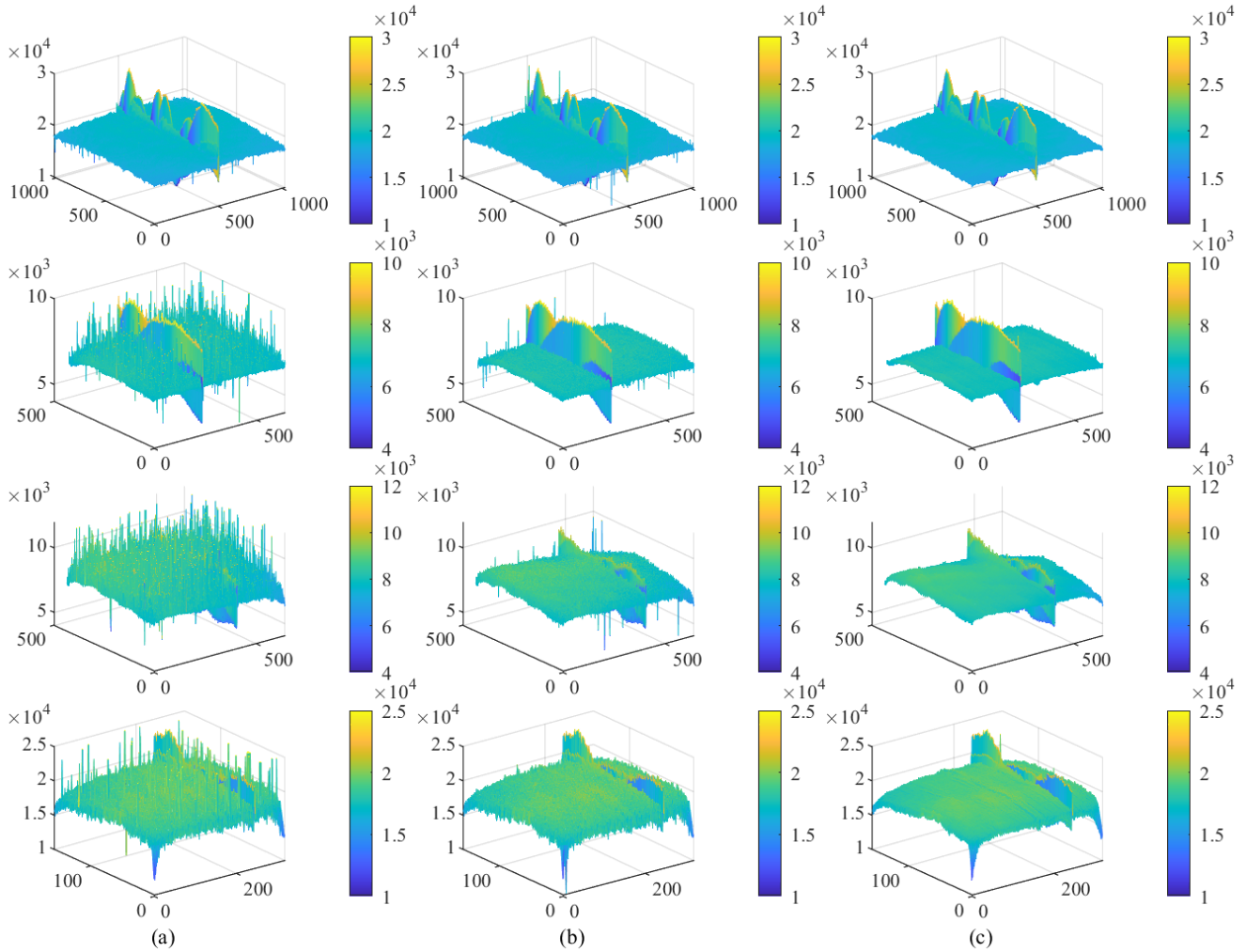



Fig. 10. (Top to Bottom) In order, 3-D visual quality comparison of the defective pixel removal results on $O_2$, $CO_2$-1, $CH_4$, and $CO_2$-2 bands, respectively. (a) Original. (b) BEAC. (c) Proposed method.

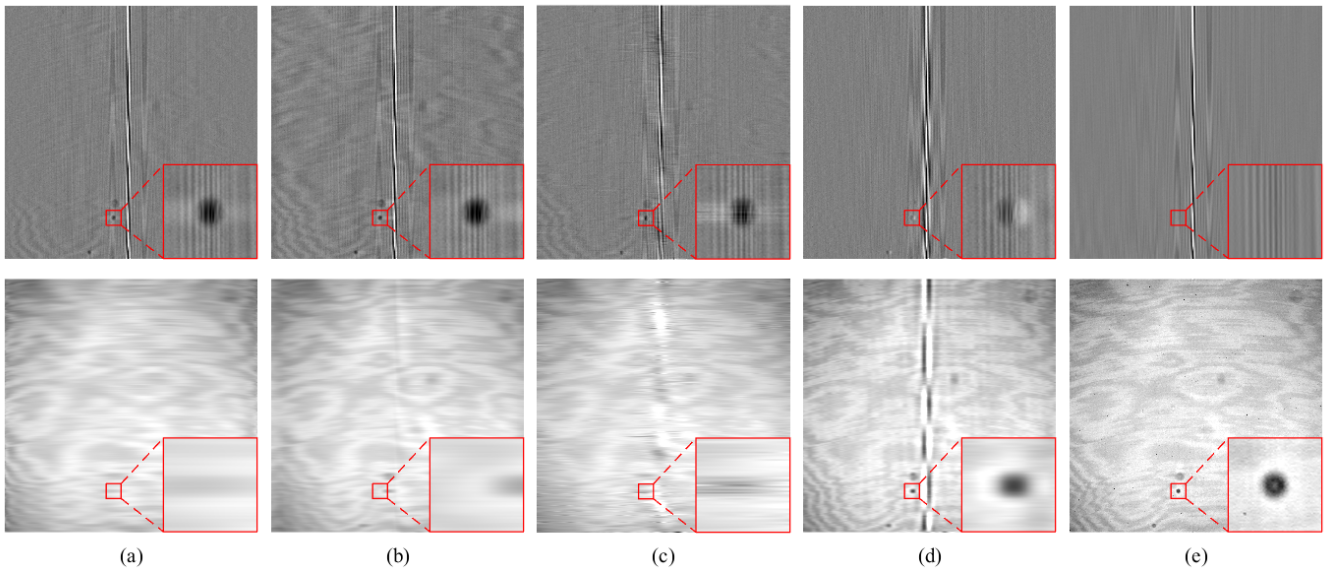



Fig. 11. Visual comparison of baseline correction results (top) and residual baseline components (bottom) on $O_2$ band interferogram image. (a) PF. (b) BLPF. (c) EMD. (d) WT. (e) Proposed method.

Fig. 16 presents the theoretical spectrum and calibrated spectra of $O_2$ band using different methods. The theoretical spectrum is simulated using the SCIATRAN radiative transfer model with the input of nadir footprint geolocation, solar-viewing geometry, environmental parameters (surface reflectance, aerosol, etc.) and then convolved with

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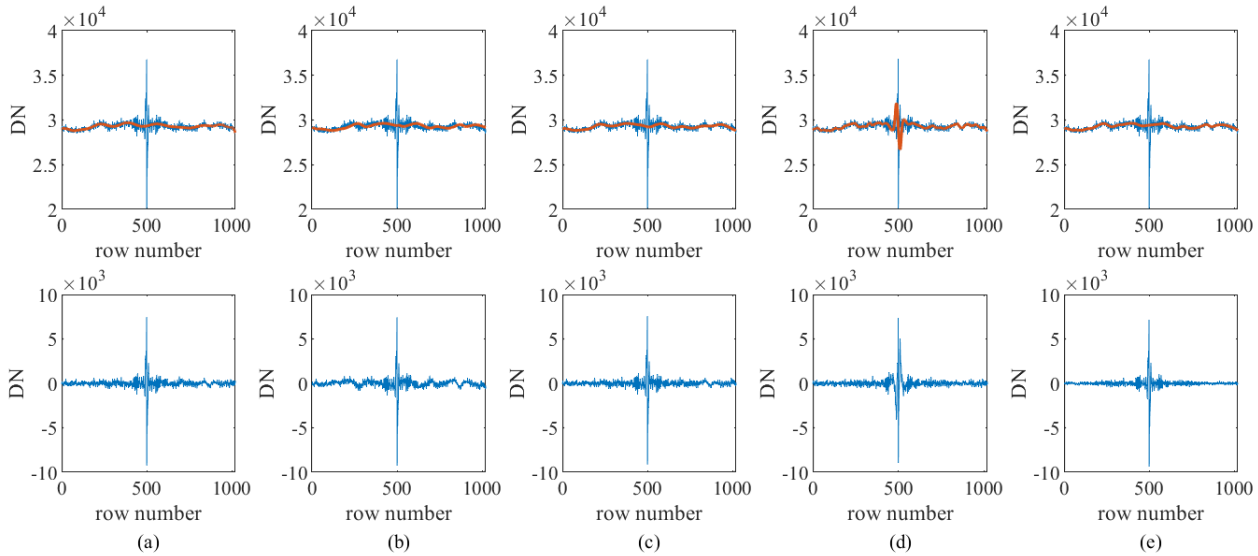



Fig. 12. The 500th row of the original interferogram with the fit baseline delineated in a red curve (top) and the baseline correction result using different methods (bottom). (a) PF. (b) BLPF. (c) EMD. (d) WT. (e) Proposed method.

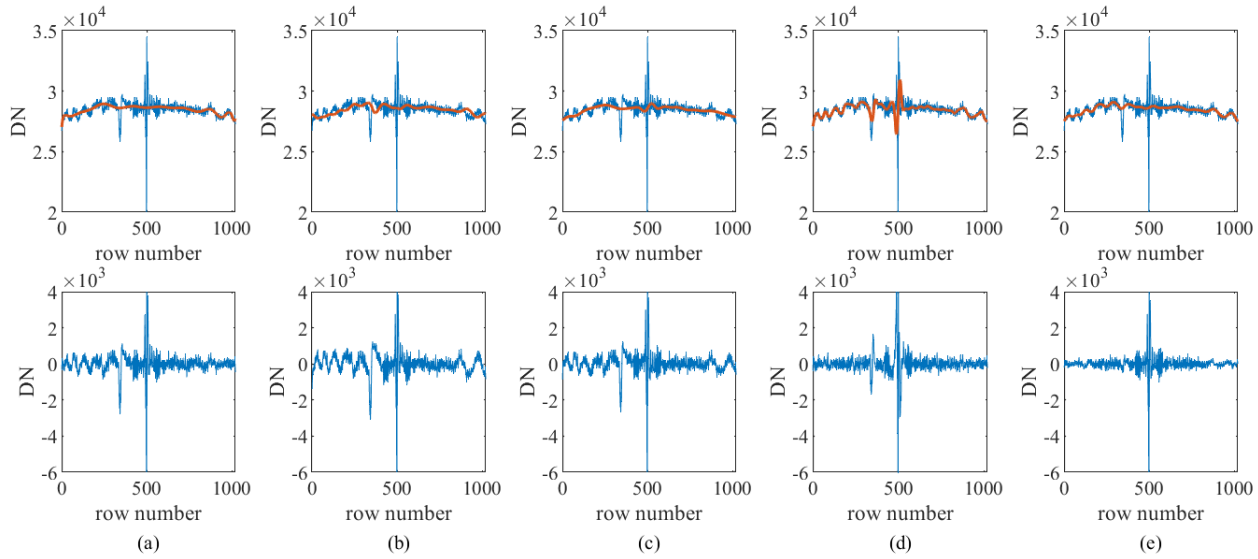



Fig. 13. The 996th row of the original interferogram with the fit baseline delineated in a red curve (top) and the baseline correction result using different methods (bottom). (a) PF. (b) BLPF. (c) EMD. (d) WT. (e) Proposed method.

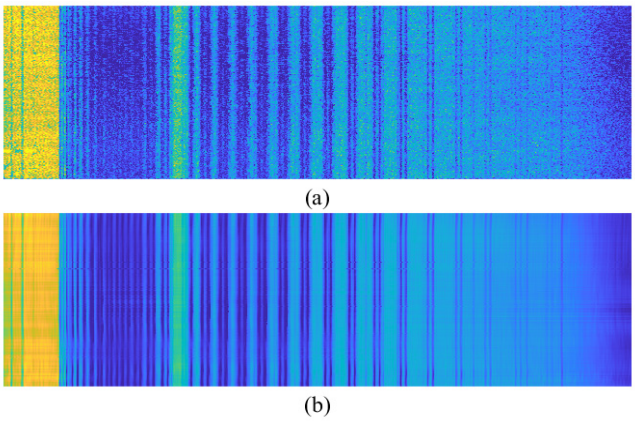



Fig. 14. Spectrogram of the interferogram corrected by different methods. (a) BEAC + PF method. (b) Proposed method.

the instrumental line shape (ILS) function of GMI measured in the laboratory [28], [29]. The calibrated spectra are calculated by row-wise averaging and spectra/radiometric calibration. Compared with other competing methods, the proposed method yields the most similar spectrum with the theoretical spectrum.

To quantitatively evaluate the performance of the proposed method on the recovered spectra, the mean standard deviation (MSTD) metric and spectral angle mapper (SAM) metric are adopted. MSTD metric describes the dispersion of spectrogram, and SAM metric measures the similarity between calibrated spectrum and theoretical spectrum. The two metrics are formulated as

$$\mathrm{MSTD} = \frac{1}{N}\sum_{j=1}^{N} \mathrm{std}\left(\mathbf{R}_j\right) \tag{33}$$

$$\mathrm{SAM} = \frac{180}{\pi} \times \arccos\left(\frac{\boldsymbol{r}_{\mathrm{calib}} \cdot \boldsymbol{r}_{\mathrm{theor}}}{|\boldsymbol{r}_{\mathrm{calib}}||\boldsymbol{r}_{\mathrm{theor}}|}\right) \tag{34}$$

where $\mathbf{R}_j$ denotes the $j$th column of the spectrogram; $\boldsymbol{r}_{\mathrm{calib}}$ and $\boldsymbol{r}_{\mathrm{theor}}$ are the calibrated spectrum and theoretical spectrum, respectively. Note that, the theoretical spectrum

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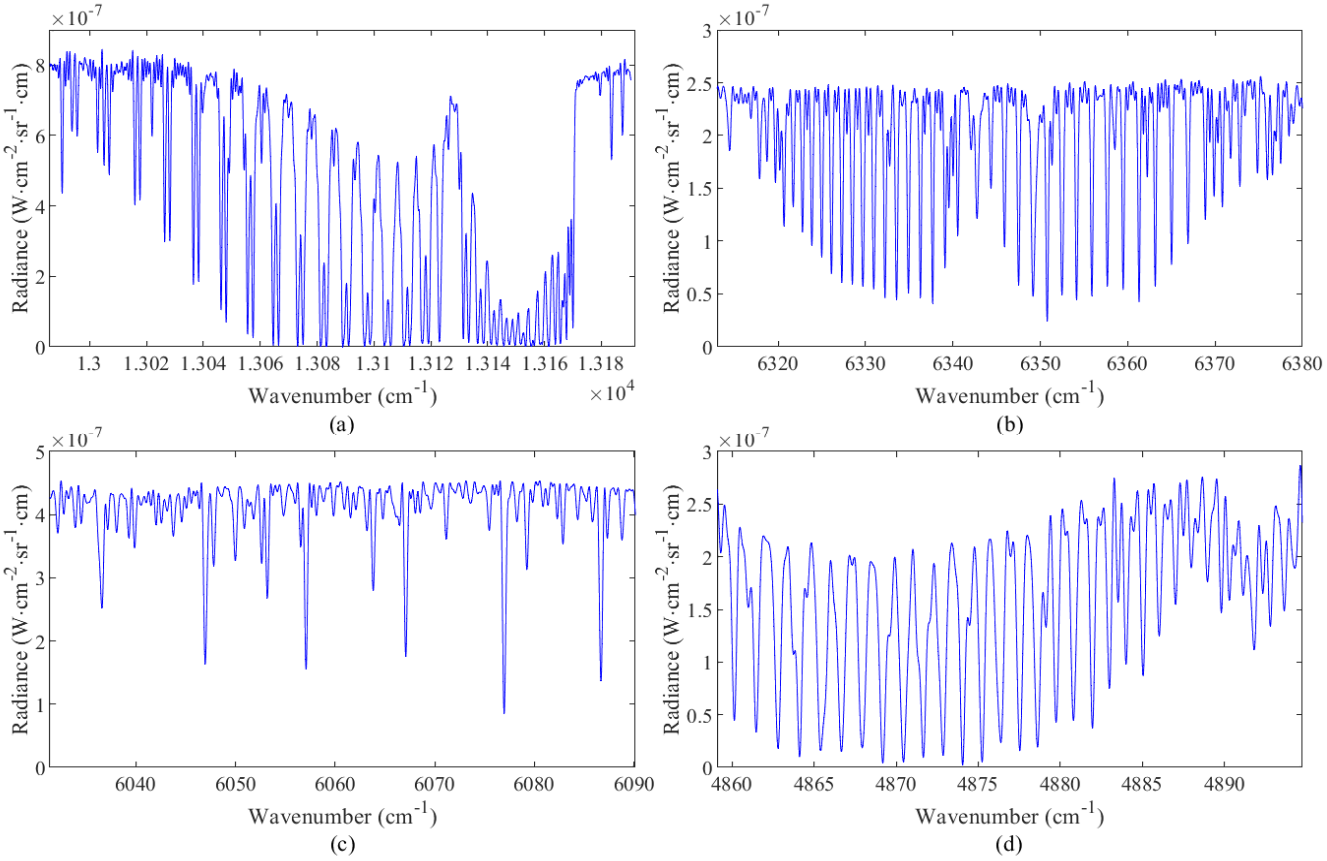



Fig. 15. Calibrated spectra of corrected interferogram image using the proposed method. (a) $O_2$ band. (b) $CO_2$-1 band. (c) $CH_4$ band. (d) $CO_2$-2 band.

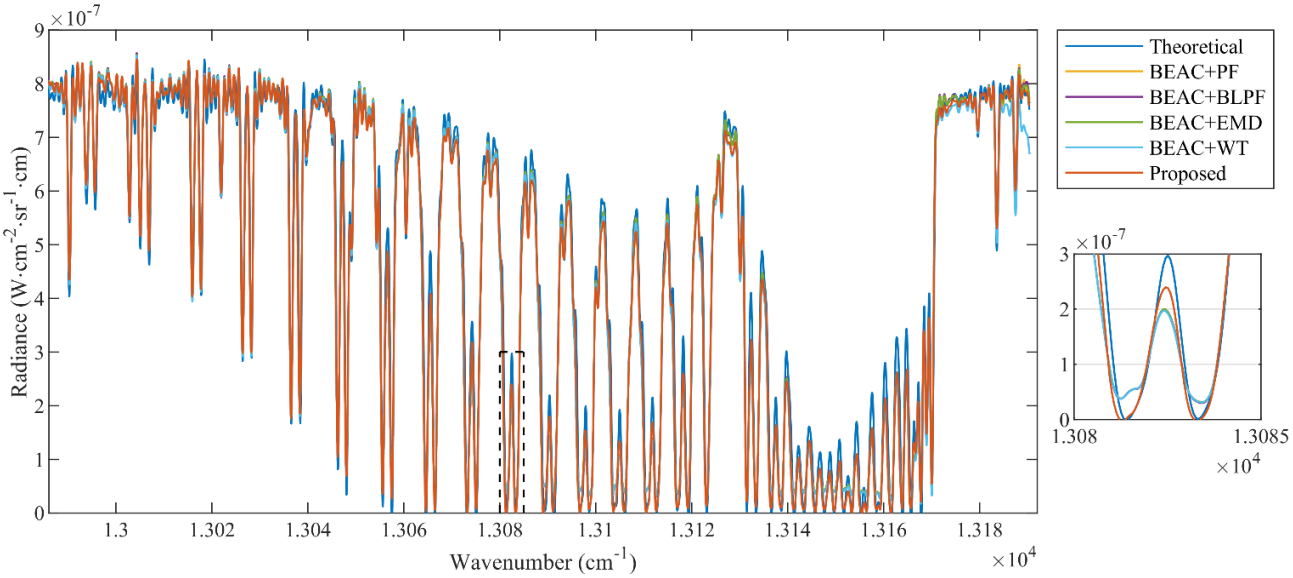



Fig. 16. Calibrated spectra superimposed on the theoretical spectrum. The plots in the doted box are magnified and shown on the bottom right.

should be interpolated into the wavenumber grid of calibrated spectrum.

Generally, the smaller value of MSTD and SAM signifies that the performance of denoising is better.

The quantitative evaluation results of four bands are shown in Table II. The best results for each metric are labeled in bold, and the second-best results are underlined. Here, it is clear that the proposed method provides the smallest values in both MSTD and SAM for all four bands, which confirms the advantage of the proposed method over the other competing methods.

### D. Discussions

*1) Sensitivity Analysis of Parameters:* In the proposed model, there are several parameters that need to be carefully identified, i.e., the desired rank $r$, the regularization

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TABLE II
QUANTITATIVE EVALUATION OF DIFFERENT METHODS

| Band | Evaluation index | BEAC+PF | BEAC+BLPF | BEAC+EMD | BEAC+WT | Proposed |
|---|---|---|---|---|---|---|
| $O_2$ | MSTD | $4.307\times10^3$ | $4.326\times10^3$ | $4.312\times10^3$ | $4.302\times10^3$ | **$1.084\times10^3$** |
| | SAM (°) | 3.698 | 3.709 | 3.693 | 3.702 | **3.198** |
| $CO_2$-1 | MSTD | 601.786 | 602.273 | 602.134 | 602.247 | **128.898** |
| | SAM (°) | 4.328 | 4.345 | 4.337 | 4.343 | **3.429** |
| $CH_4$ | MSTD | 701.328 | 726.681 | 702.896 | 701.319 | **232.214** |
| | SAM (°) | 4.560 | 4.856 | 4.672 | 4.626 | **4.534** |
| $CO_2$-2 | MSTD | 714.611 | 724.169 | 716.011 | 716.127 | **572.056** |
| | SAM (°) | 3.828 | 3.923 | 3.879 | 3.903 | **3.344** |
| Total Time Cost (s) | | **0.65** | 1.3 | 5.8 | 22.5 | 8.9 |

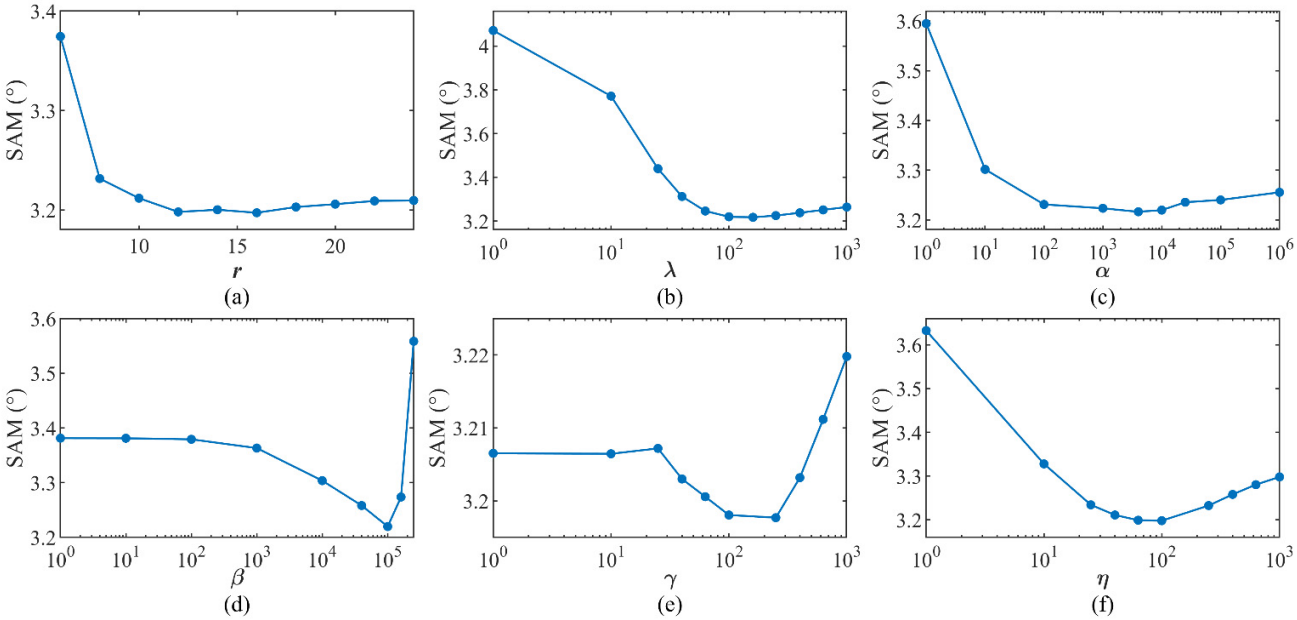



Fig. 17. Sensitivity analysis of parameters involved in the proposed method. (a) SAM values with different $r$. (b) SAM values with different $\lambda$. (c) SAM values with different $\alpha$. (d) SAM values with different $\beta$. (e) SAM values with different $\gamma$. (f) SAM values with different $\eta$.

parameters $\lambda$, $\alpha$, $\beta$, $\gamma$, and $\eta$. To search for the optimal values of these parameters, the change in SAM is used as the criterion for parameter setting. Take $O_2$ band as an example, the performance of the proposed method in terms of SAM with different parameter settings is shown in Fig. 17.

The rank $r$ embodies the nonlocal self-similarity of the interferogram image. From Fig. 17(a), we can see that the SAM values of the proposed method are robustness with the values of $r$, when $r > 12$. Considering both denoising performance and computation efficiency, the rank is empirically set to 12.

The parameter $\lambda$ determines the sparsity of the defective pixels. From Fig. 17(b), it can be seen the smallest SAM value achieves at $\lambda = 10^2$.

The parameters $\alpha$ and $\beta$ involved in the PLS mode promote the smoothness and fidelity of baseline, respectively. As shown in Figs. 15(d) and 17(c), the SAM values versus parameter $\alpha$ remain almost stable in the range of $10^2$–$10^5$. On the contrary, the SAM values are more sensible to the parameter $\beta$, and the smallest value is achieved at $\beta = 10^5$.

The parameters $\gamma$ and $\eta$ are combined to restrict the unidirectional property of the interferogram. As shown in Fig. 17(e) and (f), the smallest SAM values are achieved at $\gamma = \eta = 10^2$. The parameter $\gamma$, by contrast, has a small impact on the performance of the proposed method.

*2) Convergence Analysis:* To verify the convergence of the proposed method, we give the SAM values versus iteration number in Fig. 18. We can see that as the iteration progresses, the SAM values decrease monotonically, indicating the performance improvement of the proposed method. Meanwhile, the SAM value decreases rapidly in the first few iterations and then grows slowly in the subsequent iterations. After >15 iterations, the SAM value no longer shows any significant change, which indicates that the proposed method has a high convergent speed.

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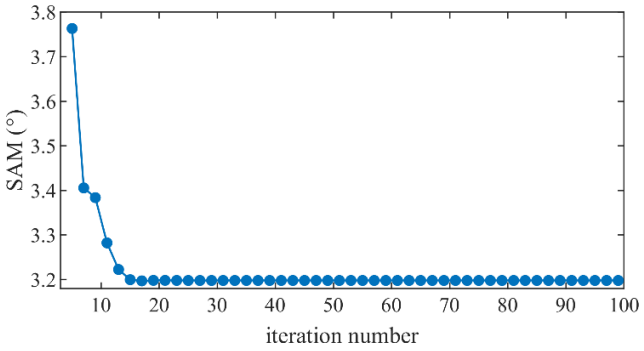



Fig. 18. SAM values versus iteration number.

*3) Computational Time Analysis:* Despite the higher computational complexity, the time cost of the proposed method remains within the acceptable range for real-time processing requirements and is even lower than that of the BEAC + WT method (refer to Table II). This can be attributed to 1) its fast convergence and 2) more efficient matrix operations compared to the loop-based operations. It is worth noting that all algorithms are currently computed on the CPU. Therefore, the computation time of the proposed method can be further reduced by implementing GPU acceleration.

## V. Conclusion

In this article, we have proposed a mixed noise removal method for GMI interferograms by combining the RPCA, UTV, and PLS models into a unified optimization framework. The proposed method can simultaneously remove defective pixels, baseline drifts and random noise from the measured interferograms, while preserving the interference fringes well.

Compared with existing techniques, the main contributions of the proposed method are summarized as follows.

1) The measured interferogram is decomposed into desired interferogram, defective pixels, baseline drifts, and random noises, each of which is modeled based on its unique characteristic. To the best of our knowledge, the proposed method represents a pioneering approach in handling the mixed noise simultaneously while avoiding coupling and interaction.
2) The ADMM framework has been employed to address the optimization problem efficiently, rendering the proposed method suitable for practical engineering applications.
3) The parameters of the proposed method have been adjusted carefully to ensure the recovered spectra closest to the theoretical spectra.

The application of the proposed method in the level-1 data processing of GMI payload will contribute to improve the quality of recovered spectra, and thus increase the retrieval accuracy of GHGs. In addition, the secondary products of the proposed method, i.e., the defective pixels and baseline drifts, can be fully utilized to monitor the changes in the instrument performance.

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